# 🧠 Chain-of-Thought Prompting

**Day 2 — Notebook 3 of 5 | Estimated Time: 25 minutes**

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## What You'll Learn

- What chain-of-thought (CoT) prompting is and why it improves reasoning
- How to use CoT for math, logic, and multi-step problems
- The "Let's think step by step" technique
- When CoT helps vs when it's unnecessary

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## What is Chain-of-Thought Prompting?

**Chain-of-thought (CoT)** prompting encourages the model to break down its reasoning into **explicit steps** before giving a final answer.

Without CoT, models often jump to answers — and get them wrong on complex tasks. With CoT, accuracy improves significantly on:

- Math and arithmetic problems
- Logic puzzles
- Multi-step reasoning tasks
- Analytical questions

## Setup

```
%pip install -U -q "google-genai>=1.0.0"
```

```
!pip install google-auth==2.47.0
```

```
Requirement already satisfied: google-auth==2.47.0 in /usr/local/lib/python3.12/dist-packages (2.47.0)
Requirement already satisfied: pyasn1-modules>=0.2.1 in /usr/local/lib/python3.12/dist-packages (from google-auth==2.47.0) (0.4.
Requirement already satisfied: rsa<5,>=3.1.4 in /usr/local/lib/python3.12/dist-packages (from google-auth==2.47.0) (4.9.1)
Requirement already satisfied: pyasn1<0.7.0,>=0.6.1 in /usr/local/lib/python3.12/dist-packages (from pyasn1-modules>=0.2.1->goog
```

```
import sys, os
import time

import json
import re
from google import genai
from google.genai import types
from IPython.display import Markdown


try:
    from google.colab import auth
    auth.authenticate_user()
    print("✅ Authenticated via Colab")
except ImportError:
    print("✅ Using local credentials (gcloud auth)")


client = genai.Client(
    vertexai=True,
    project="chanakya-476010",
    location="us-central1",
)
MODEL_ID = "gemini-2.5-flash"
print("✅ Ready!")
```

```
✅ Authenticated via Colab
✅ Ready!
```

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## 1. The Problem: Without Chain-of-Thought

Let's see what happens when we ask a multi-step math problem without CoT:

```
# Without CoT
prompt_no_cot = """
A store sells apples for $2 each. They have a promotion: buy 3, get 1 free.
How much would 10 apples cost?
"""

response = client.models.generate_content(model=MODEL_ID, contents=prompt_no_cot)
print("Without CoT:")
print(response.text)
```

```
Without CoT:
Let's break this down:

1.  **Understand the promotion:** For every 3 apples you *buy*, you get 1 free. This means you get 4 apples for the price of 3.
    *   Cost of 3 apples = 3 * $2 = $6
    *   So, you get 4 apples for $6.

2.  **Calculate for 10 apples:**
    *   You need 10 apples.
    *   You can get two sets of the "buy 3, get 1 free" deal.
        *   Set 1: Buy 3, get 1 free = 4 apples for $6.
        *   Set 2: Buy 3, get 1 free = 4 apples for $6.
    *   So far, you have 8 apples (4+4) for a total cost of $12 ($6+$6).

3.  **Remaining apples:**
    *   You need 10 apples, and you have 8.
    *   You still need 2 more apples (10 - 8 = 2).
    *   These 2 apples are not enough to trigger another "buy 3, get 1 free" deal, so you buy them at the regular price.
    *   Cost of 2 additional apples = 2 * $2 = $4.

4.  **Total Cost:**
    *   Cost of the first 8 apples (with promotion) = $12
    *   Cost of the remaining 2 apples (regular price) = $4
    *   Total cost = $12 + $4 = $16

10 apples would cost **$16**.
```

```
# With CoT — just add "Think step by step"
prompt_cot = """
A store sells apples for $2 each. They have a promotion: buy 3, get 1 free.
How much would 10 apples cost?

Think step by step.
"""

response = client.models.generate_content(model=MODEL_ID, contents=prompt_cot)
print("With CoT:")
Markdown(response.text)
```

```
With CoT:
```

Here's how to break down the cost step-by-step:

1. **Understand the Promotion Block:**
   - You buy 3 apples, and get 1 free.
   - This means for every *4 apples* you get, you only pay for *3 of them*.
   - Cost of one block (4 apples) = 3 apples * $2/apple =$6.
2. **Determine How Many Full Promotion Blocks You Can Get:**
   - You need 10 apples.
   - Each block gives you 4 apples.
   - 10 apples / 4 apples/block = 2 full blocks with a remainder of 2 apples.
3. **Calculate the Cost for the Full Blocks:**
   - You have 2 full blocks, and each block costs $6. * Cost for 8 apples (2 blocks) = 2*$6 = $12.
4. **Calculate the Cost for the Remaining Apples:**
   - You needed 10 apples, and you've accounted for 8.
   - Remaining apples = 10 - 8 = 2 apples.
   - These 2 apples cannot form a new "buy 3 get 1 free" deal, so they must be bought at the regular price.
   - Cost for 2 apples = 2 apples * $2/apple =$4.
5. **Calculate the Total Cost:**
   - Total cost = Cost of full blocks + Cost of remaining apples
   - Total cost = $12 + $4 = $16.

10 apples would cost **$16**.

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## 2. Zero-Shot CoT: "Let's think step by step"

The simplest form of CoT is called **zero-shot CoT** — just adding the phrase *"Let's think step by step"* to any prompt.

```
# Logic puzzle
prompt = """
There are 5 people in a room. Each person shakes hands with every other person exactly once.
How many handshakes occur in total?

Let's think step by step.
"""

response = client.models.generate_content(model=MODEL_ID, contents=prompt)
Markdown(response.text)
```

Let's break this down step-by-step:

1. **Consider the first person:** This person shakes hands with the other 4 people. (4 handshakes)
2. **Consider the second person:** This person has already shaken hands with the first person. So, they need to shake hands with the remaining 3 people. (3 new handshakes)
3. **Consider the third person:** This person has already shaken hands with the first two. So, they need to shake hands with the remaining 2 people. (2 new handshakes)
4. **Consider the fourth person:** This person has already shaken hands with the first three. So, they need to shake hands with the remaining 1 person. (1 new handshake)
5. **Consider the fifth person:** This person has already shaken hands with all the previous four people. They have no new handshakes to make. (0 new handshakes)

**Total handshakes:** $4 + 3 + 2 + 1 + 0 = 10$

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**Alternative (more mathematical) approach:**

- Each person shakes hands with every other person.
- There are 5 people. Each person shakes hands with $(5 - 1) = 4$ other people.
- If we multiply $5 \times 4 = 20$, we are counting each handshake twice (e.g., Person A shaking Person B's hand is counted once for A and once for B).
- Therefore, we need to divide the total by 2: $20/2 = 10$.

This is a classic problem that can be solved using combinations. We are choosing 2 people out of 5 to form a handshake, and the order doesn't matter (A-B is the same as B-A). The formula for combinations "n choose k" is C(n, k) = n! / (k! * (n-k)!), where n is the total number of items, and k is the number of items to choose.

Here, n=5 (people) and k=2 (people per handshake): C(5, 2) = 5! / (2! * (5-2)!) = 5! / (2! * 3!) = (5 * 4 * 3 * 2 * 1) / ((2 * 1) * (3 * 2 * 1)) = (5 * 4) / (2 * 1) = 20 / 2 = 10

```
# Word problem
prompt = """
Sarah has 3 times as many books as Tom. Tom has 5 more books than Lisa.
Lisa has 8 books. How many books does Sarah have?

Let's think step by step.
```

```
"""

response = client.models.generate_content(model=MODEL_ID, contents=prompt)
Markdown(response.text)
```

Let's break this down step by step:

1. **Find out how many books Lisa has:** The problem states: Lisa has 8 books.
2. **Find out how many books Tom has:** The problem states: Tom has 5 more books than Lisa. So, Tom has 8 + 5 = 13 books.
3. **Find out how many books Sarah has:** The problem states: Sarah has 3 times as many books as Tom. So, Sarah has 3 * 13 = 39 books.

**Answer:** Sarah has 39 books.

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## 3. Few-Shot CoT: Showing the Reasoning Pattern

For even better results, provide examples **with reasoning steps included**.

```
prompt = """
Solve the following problems by showing your reasoning step by step.

Problem: If a train travels 60 mph for 2.5 hours, how far does it go?
Reasoning:
- Speed = 60 mph
- Time = 2.5 hours
- Distance = Speed × Time = 60 × 2.5 = 150 miles
Answer: 150 miles

Problem: A recipe calls for 2 cups of flour for 12 cookies. How much flour for 30 cookies?
Reasoning:
- 12 cookies need 2 cups
- 1 cookie needs 2/12 = 1/6 cup
- 30 cookies need 30 × 1/6 = 5 cups
Answer: 5 cups

Problem: A rectangular garden is 15 meters long and 8 meters wide. A path that is 1 meter wide goes around the entire garden. Wh
Reasoning:
"""

response = client.models.generate_content(model=MODEL_ID, contents=prompt)
Markdown(response.text)
```

- Garden length = 15 meters
- Garden width = 8 meters
- Path width = 1 meter
- Area of the garden = Length × Width = 15 × 8 = 120 square meters
- The path goes around the garden, adding 1 meter to each side.
- New length (garden + path) = Garden length + 2 × Path width = 15 + 2 × 1 = 15 + 2 = 17 meters
- New width (garden + path) = Garden width + 2 × Path width = 8 + 2 × 1 = 8 + 2 = 10 meters
- Area of the garden including the path = New length × New width = 17 × 10 = 170 square meters
- Area of just the path = (Area of garden + path) - (Area of garden) = 170 - 120 = 50 square meters Answer: 50 square meters

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## 4. CoT for Business Analysis

CoT isn't just for math — it's great for any analytical task.

```
prompt = """
Analyze whether this company should expand into the European market.

Context:
- Company: SaaS startup, $10M annual revenue
- Current markets: US, Canada
- European competitors: 3 established players
- Company growth rate: 40% YoY
- Available budget for expansion: $2M

Think through this step by step, considering:
1. Market opportunity
```

```
2. Competitive landscape
3. Financial readiness
4. Risks
5. Final recommendation
"""

response = client.models.generate_content(model=MODEL_ID, contents=prompt)
Markdown(response.text)
```

### Analysis: European Market Expansion for SaaS Startup

**Context Recap:**

- **Company:** SaaS startup
- **Revenue:** $10MARR***CurrentMarkets:**US,Canada***EuropeanCompetitors:**3establishedplayers***CompanyGrowthRate:**40***AvailableBudgetforExpansion:**$2M

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#### 1. Market Opportunity

**Positive Aspects:**

- **Vast and Diverse Market:** Europe represents a massive economic bloc with over 740 million people and a collective GDP comparable to or exceeding the US. It offers a huge untapped customer base for a successful SaaS product.
- **High SaaS Adoption:** Western European countries, particularly the Nordics, UK, Germany, and France, have a high degree of digital maturity and a strong appetite for SaaS solutions across various industries. SaaS adoption rates are steadily growing.
- **Recurring Revenue Potential:** A successful expansion would significantly increase the company's total addressable market (TAM) and, consequently, its potential for recurring revenue, bolstering its valuation and long-term stability.
- **Geographic Diversification:** Reduces reliance on a single economic region (North America), mitigating risks associated with regional downturns or market saturation.
- **Potential for Underserved Niches:** Despite established competitors, the vastness of Europe might mean there are specific industries, business sizes (e.g., SMBs vs. Enterprise), or even particular countries where the established players are weaker or less focused.
- **Follow-the-Customer Strategy:** If current US/Canada enterprise customers have European operations, expanding into Europe could serve existing customer needs and deepen relationships.

**Considerations/Unknowns:**

- **Market Fragmentation:** Europe is not a single market. It's a collection of distinct countries with different languages, cultures, business practices, and legal frameworks. A "European" strategy often needs to be a "country-by-country" strategy.
- **Product-Market Fit in Europe:** While successful in North America, it's not guaranteed the product will resonate equally well in all European markets without adaptation.
- **Regulatory Differences:** GDPR is a major factor for any SaaS company dealing with European data, which needs strict adherence. Other local regulations (e.g., e-invoicing standards, data localization) can also apply.

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#### 2. Competitive Landscape

**Challenges:**

- **Established Incumbents:** Having "3 established players" means significant barriers to entry. These competitors likely have:
  - Strong brand recognition and trust.
  - Existing customer bases and long-term contracts.
  - Localized products, support, and sales teams.
  - Deep understanding of local market nuances and regulations.
  - Potentially more aggressive pricing or bundled solutions.
- **Price Sensitivity:** Some European markets can be more price-sensitive than the US, especially for SMBs, which could challenge the company's existing pricing model.
- **Switching Costs:** If the competitors' products are deeply embedded into customer workflows, switching to a new provider (even a superior one) can be a high hurdle.

**Opportunities for Differentiation:**

- **Niche Focus:** Can the company target a specific vertical or segment within Europe that the established players are not adequately serving?
- **Superior Product/Technology:** Is the company's SaaS genuinely more innovative, user-friendly, or technologically advanced? A 40% YoY growth suggests strong product-market fit in its current regions.
- **Exceptional Customer Service:** US companies often differentiate on customer service; maintaining this high standard with localized support can be a strong advantage.
- **Pricing Strategy:** A disruptive pricing model could attract customers, but needs careful consideration not to devalue the product.
- **Modern UX/UI:** Newer SaaS products often boast more intuitive interfaces compared to older, established systems.
- **Compliance as a Feature:** If the company can be unequivocally GDPR-compliant and potentially offer features that help customers meet their own regulatory requirements, it can be a strong selling point.

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#### 3. Financial Readiness

**Positive Aspects:**

- **Healthy Growth Rate (40% YoY):** This indicates strong product-market fit and efficient customer acquisition in current markets. It suggests the company is well-managed and its core business is thriving, providing a solid foundation.
- **Existing Revenue** ($10MARR$)$: ** Thisisasubstantialbase, providingcredibilityandpotentiallyinternalcashflowtosupporttheexpansionbeyondtheallocatedbudge$ $** DedicatedExpansionBudget($**2M):** This is a significant sum, indicating a serious commitment to the expansion. It's enough to fund a strategic, phased entry rather than a tentative dip.

- **Burn Rate:** $econsumedquicklybyinternationalexpansioncosts(hiringlocaltalent, legalfees, localization, mark$
$** Capital Allocation: ** How will the$ 2M be distributed?
  - ...try manager, sales team, support staff (major cost).
  - **Localization:** Product, website, marketing materials, legal docs.
  - **Legal & Compliance:** Setting up entities, GDPR counsel.
  - **Marketing & Sales:** Initial campaigns, lead generation.
  - **Infrastructure:** CRM, support tools, potential data center costs.
- **Opportunity Cost:** What could $2M do for the company's growth in existing markets (e.g., deeper penetration, R&D for new features, larger US/Canada sales team)? Is Europe the *best* use of this capital for accelerating growth?
- **Time to Profitability:** International expansion rarely generates immediate returns. The company needs to be prepared for potentially 18-36 months of investment before seeing significant profits from the new market.

## 5. When to Use CoT

| Use CoT ✅ | Skip CoT ❌ |
|---|---|
| Math / arithmetic | Simple factual questions |
| Logic puzzles | Text translation |
| Multi-step reasoning | Sentiment analysis |
| Decision making | Creative writing |
| Code debugging | Simple classification |